

Vinith Ramaswamy

Software Engineer · Founder of JustBook · UW–Madison CS & Data Science

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SUMMARY

Software engineer who has taken a product from idea to production alone: founder and sole engineer of JustBook, a multi-tenant booking and payments platform live on web, iOS and Android (Python, Flask, PostgreSQL, React, Stripe). Also built ride-dispatch, Go microservices on Kubernetes with gRPC, RabbitMQ and Kafka.

EDUCATION

B.S. Computer Science & B.S. Data Science GPA 3.96 / 4.0

Aug 2023 – May 2026

University of Wisconsin–Madison · *Graduated with Distinction, in three years · Certificate in Entrepreneurship*

Coursework: Big Data Systems, Honors Algorithms, Databases, AI, Machine Learning, Machine Organization

TECHNICAL SKILLS

Languages: Python, Go, SQL, JavaScript, TypeScript, Java, C++, C, R

Backend & data: Flask, SQLAlchemy, PostgreSQL, REST API design, gRPC, Redis, MongoDB, Kafka, RabbitMQ, Spark

Infrastructure: Docker, Kubernetes, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Render, Cloudflare Workers, AWS S3

Frontend & mobile: React 19, Next.js, Vite, Capacitor (iOS/Android), React Native, Figma

Testing: pytest, Vitest, Playwright, k6 load testing

Payments & security: Stripe (Connect, Billing, Terminal, Tap to Pay), Twilio, JWT, TOTP, WebAuthn passkeys, RBAC

EXPERIENCE

Founder & Software Engineer

May 2025 – Present

JustBook · *Booking and payments SaaS* · justbookapp.com · [App Store](#) · [Google Play](#) · [Write-up](#)

- **Built and shipped a multi-tenant booking and payments platform alone in 15 months**, live on web, iOS and Android: Flask/PostgreSQL API (633 endpoints, 127 models), React 19 front end, and Capacitor mobile apps with Stripe Tap to Pay.
- **Designed the availability engine** (time zones, DST, rosters, holds) shared by the booking site, admin calendar and waitlist so they cannot disagree; a constant 5 queries per month view, pinned by query-budget tests; published on GitHub.
- **Eliminated double bookings under concurrency**: reproduced the race with a k6 load test, then serialized all five booking write paths with PostgreSQL advisory locks keyed per branch and day, ordered to prevent deadlock.
- **Integrated Stripe Connect, Billing and Terminal** (card readers, Tap to Pay on iPhone and Android) with replay-safe webhooks: exactly-once guards inside each money helper, backed by partial unique indexes.
- **Cut the booking-flow read from nine sequential queries to one** on warm hits (about 0.8 s cold before) with request-scoped memoization and a commit-invalidated TTL cache; query-budget tests fail the build on regressions.
- **Enforced tenant isolation mechanically**: an AST-based audit fails the test suite on any unguarded route, and a 181-code error contract generates the frontend constants; JWT with revocation, TOTP, passkeys and RBAC.

Web Developer & Product Consultant

Mar 2026 – May 2026

NeuroSolis Therapeutics · *Contract · Remote*

- **Designed, built and deployed the investor-facing website** with the founding team, from content structure and visual direction through domain configuration, turning complex biotech concepts into clear copy.

IT Help Desk Technician, Level 1 to Lead

Jan 2024 – May 2026

UW–Madison Libraries · *Part-time · Madison, WI*

- **Promoted through four levels to Lead within a year**; led daily operations across public computer labs and the help desk (Active Directory, Bomgar, InfoBlox, Redmine), 29 hours a week alongside a full-time dual degree.

PROJECTS

ride-dispatch Ride-sharing backend in Go microservices

2026

github.com/vramaswamy4/ride-dispatch · [Write-up](#)

- **Built a ride-sharing backend as five Go microservices on Kubernetes**, using gRPC where the caller needs an answer, RabbitMQ topic exchanges for events and commands, and WebSockets for live rider and driver updates.
- **Designed a real-time driver-location pipeline**: Kafka partitioned by driver ID into an idempotent consumer that maintains a Redis geo index, making nearest-driver matching one GEOSEARCH and the driver service stateless.
- **Made message handling crash-safe** with durable queues, manual acks, backoff retries, dead-lettering and idempotent handlers, proven by a chaos test that kills a pod mid-ride and asserts redelivery and no double-processing.